DEPI • Professional Data Analyst Track

Data-Driven Healthcare Insights

Project Documentation — Part 1

Proposed Topic: Analysis of Factors Associated with Mortality Risk in Egypt

Team: Team3

Document status	Submission target
Draft for team review; project topic not yet approved by the Technical Instructor.	16 October 2026

Important data note: The supplied file named Egypt_Synthetic_Mortality_Risk_250000.csv is synthetic, not an official Egyptian mortality dataset. It must not be presented as real Egyptian mortality records or used to claim actual national mortality rates. The project will use verified official sources for real-world findings.

1. Project Planning & Management

1.1 Project Proposal

Project overview. The project proposes a data-analysis workflow to explore how selected demographic, health, lifestyle, and healthcare-access indicators are associated with mortality-related measures in Egypt. The analysis will rely on verified public data from official sources, subject to confirmation of country coverage, years, definitions, and data completeness.

Problem statement (proposed). Public health decisions require interpretable evidence about mortality patterns and the factors associated with them. However, mortality data and health indicators may come from different sources, years, and measurement systems. The project will assess what can be reliably compared, identify observable patterns, and communicate limitations alongside findings.

Proposed solution. Collect and document suitable official datasets; clean and harmonize indicators; perform exploratory and statistical analysis using Python and SQL; and build an interactive Power BI or Tableau dashboard that presents trends, comparisons, and caveats.

Scope. The initial scope is Egypt at national level, using the years and demographic breakdowns supported by the selected official datasets. Governorate-level or individual-level analyses will be included only if verified public data support them.

Out of scope. The project will not diagnose individuals, provide clinical recommendations, claim causal effects from correlation alone, or infer real Egyptian mortality rates from synthetic records.

1.2 Objectives

- Identify and document official, accessible mortality and health-indicator datasets relevant to Egypt.
- Assess coverage, definitions, years, missingness, and comparability before combining data.
- Explore trends and differences by cause of death, age, sex, and other dimensions where data permit.
- Examine associations between mortality measures and selected health or socioeconomic indicators without overstating causality.
- Deliver a reproducible data-preparation and analysis workflow plus an interactive dashboard and documented findings.

1.3 Success KPIs (proposed)

KPI	Target / definition
Source traceability	100% of final analytical fields linked to a documented source, definition, year, and URL.
Data quality	Missingness, duplicates, and consistency checks reported for every dataset.
Reproducibility	Data cleaning and analytical steps saved in version-controlled scripts/notebooks.
Dashboard coverage	Dashboard answers the agreed research questions and clearly displays source/year/limitations.
Documentation	Required project documents and role assignments maintained in GitHub.

1.4 Team Roles and Responsibilities

The following allocation is a proposed starting point and must be confirmed by all team members. Members may collaborate across tasks.

Member	Proposed responsibility
Ahmed Hamada Ahmed Abdelfadel	Team Leader; coordination and documentation; Tableau development (with Hanan).
Hanan Ibrahim Hafez Zaki	Literature review, source verification, and Tableau development (with Ahmed).
Abdullah Abdulrahman Abdulhamid Al-Hanash	Data sourcing, data dictionary, SQL development, and Power BI development.
Omar Sherif Ali Emam Bassiouni	Data cleaning, exploratory analysis, and Power BI development.
Mostafa Ahmed Mahmoud Fath El-Bab	SQL development and data preparation.
Mohamed Mahmoud Mohamed Wafy	Python analysis and implementation.

1.5 High-level Plan and Milestones

Target date	Milestone
16 Oct 2026	Submit Part 1: planning & management, literature review, requirements gathering.
By next team review	Confirm instructor approval, official datasets, research questions, and final scope.
By 6 Nov 2026	Develop system analysis/design artifacts if required by the guidelines.
By 30 Nov 2026	Implementation, source code, data processing, analysis, dashboard, and testing progress.
By 4 Dec 2026	Final presentation and testing/report deliverables, subject to instructor confirmation.

The dates above follow the supplied Project Documentation Guidelines for DEPI. The team should confirm any changes with the Technical Instructor.

1.6 Risk Assessment and Mitigation

Risk	Mitigation
Official data may not contain the desired detail for Egypt	Check country/year/age/sex/cause availability before finalizing scope; keep a documented fallback using national-level estimates.
Data definitions or years differ across sources	Maintain a data dictionary; do not merge indicators without checking units, definitions, and reference years.
Mortality registration has completeness/classification limitations	Use official metadata and include a limitations note in all conclusions.
Synthetic file is mistaken for official data	Label it synthetic in the repository and exclude it from real-world mortality estimates.
Project idea has not been approved	Present this proposal and source plan to the Technical Instructor and record approval before final commitment.
Time constraints or uneven workload	Assign owners, weekly milestones, and review deliverables during the weekly meeting.

2. Literature Review

This is an initial, source-based literature review for project planning. It summarizes the official sources identified so far; it is not a claim that the team has completed an exhaustive academic literature review.

2.1 Official sources and relevance

- **World Health Organization (WHO), Global Health Estimates.** Official source for comparable mortality and health estimates by country, age, sex, cause, and time; data availability and granularity must be checked before final selection. https://www.who.int/data/global-health-estimates/
- **WHO, Egypt country health profile.** Includes Egypt-specific mortality estimates and notes important limitations in completeness and cause-of-death assignment. https://www.who.int/data/gho/data/countries/country-details/GHO/egypt
- **WHO Mortality Database.** Country-reported mortality data by year, sex, age, and cause where data meet publication requirements; confirm Egypt coverage and usable years. https://platform.who.int/mortality/
- **World Bank Data, Egypt.** Potential source for national-level health, demographic, and socioeconomic indicators; indicator-level coverage varies. https://data.worldbank.org/country/EG

2.2 What the sources support

WHO Global Health Estimates provide comparable estimates of deaths and health outcomes by country, age, sex, cause, and time. This makes them a plausible starting point for descriptive mortality analysis and for checking which dimensions are available for Egypt.

The WHO Egypt country profile presents leading causes of death and notes that Egypt's death-registration data have low completeness and/or serious cause-of-death assignment quality issues that may affect cause-specific estimates. WHO therefore advises caution in programme evaluation and cross-country comparisons. This limitation must be visible in the project interpretation.

The WHO Mortality Database describes country-reported mortality data by year, sex, age, and cause of death, where countries report data that meet the database's publication requirements. The team must confirm Egypt's actual usable years and fields in the downloadable files before deciding whether to use it as the primary dataset.

The World Bank's Egypt data portal provides national indicators including life expectancy and demographic/economic measures. These may support contextual analysis, but an association between a national indicator and mortality does not by itself establish a causal effect.

2.3 Proposed analytical questions

- How have selected mortality indicators in Egypt changed over the years available in official sources?
- Which causes of death account for the largest estimated mortality rates, and how do patterns differ by age and sex where available?
- How do selected health or socioeconomic indicators co-vary with mortality measures over comparable years?
- How sensitive are observed trends to source choice, time coverage, and data-quality limitations?

2.4 Methodological principles

- Use age-standardized rates for comparisons where official age-standardized measures are available; do not compare raw counts as if they were rates.
- Document units, definitions, reference years, age groups, sex categories, and source for each indicator.
- Use descriptive analysis and association language unless a defensible causal design is developed and approved.
- Report uncertainty and data-quality limitations; avoid ranking regions/countries when underlying comparability is weak.
- Cite every dataset and retain its metadata and access date in the GitHub repository.

2.5 Literature-review limitations

The current review is a starting point built from official data documentation. Before finalizing the project, the team should add a small number of relevant peer-reviewed studies on mortality determinants, ecological analysis, age-standardized mortality, and data quality. Each paper must be checked and cited by the team; no study-specific results are asserted here.

3. Requirements Gathering

Requirements below are proposed for a data-analysis project and must be reviewed with the team and Technical Instructor. No stakeholder interviews or user testing have been completed yet.

3.1 Stakeholders and needs

Stakeholder	Proposed need
Project team	Reliable sources, clear definitions, repeatable workflow, manageable scope, shared GitHub repository.
Technical Instructor / DEPI evaluator	A well-defined problem, appropriate data use, transparent methodology, documented contributions, and clear presentation.
Public-health or policy audience (intended audience)	Understandable trends, comparisons, caveats, and evidence that does not overstate causality.
Dashboard viewer	Clear filters, labels, units, time periods, sources, and an explanation of limitations.

3.2 User stories / use cases

- As an evaluator, I want to see the research question, scope, source provenance, and limitations so that I can judge whether the analysis is credible.
- As a dashboard viewer, I want to filter available results by year, sex, age group, and cause of death so that I can compare patterns where source data permit.
- As a team analyst, I want a documented data dictionary and repeatable cleaning steps so that results can be reproduced.
- As a project lead, I want a clear record of tasks, milestones, risks, and owners so that team progress can be monitored.

3.3 Functional requirements

ID	Requirement
FR-01	The project must ingest datasets from verified official sources and retain source URLs and metadata.
FR-02	The workflow must profile columns, types, missing values, duplicates, units, and date/year coverage.
FR-03	The workflow must standardize field names, categories, units, and year formats where valid.
FR-04	The analysis must calculate and visualize descriptive trends and comparisons supported by the data.
FR-05	The dashboard must provide filters only for dimensions present in the selected dataset.
FR-06	Visuals must show indicator definitions, units, time range, and source.
FR-07	The repository must include the README, data dictionary, source notes, analysis code/notebooks, and documentation.
FR-08	Synthetic demonstration data, if retained, must be labelled clearly and separated from official observations.

3.4 Non-functional requirements

- Accuracy: every metric must be traceable to a source and calculation.
- Reproducibility: code, transformations, and analysis steps must be version controlled.
- Usability: charts must have readable titles, units, legends, and clear filters.
- Transparency: assumptions, missing data, limitations, and synthetic data must be disclosed.
- Privacy and ethics: use aggregated public data where possible; do not include personal or identifiable health records.
- Maintainability: use consistent file naming, modular notebooks/scripts, and a clear README.

3.5 Data requirements and data-quality controls

- Required minimum metadata: source, URL, access date, country coverage, years, unit, definition, population denominator (if applicable), age/sex breakdown, and limitations.
- Data checks: missingness, duplicate keys, invalid values, inconsistent units, mismatched years, and category harmonization.
- Join controls: combine datasets only where keys and reference periods are compatible; explain any aggregation.

- Synthetic data control: the supplied CSV is not an official source and must not be used to calculate real Egypt mortality rates.

3.6 Acceptance criteria

- The selected official dataset(s) and exact indicators are documented and their Egypt coverage is confirmed.
- The research questions can be answered using available fields and time coverage.
- All displayed metrics have definitions, units, years, and source citations.
- The dashboard conclusions match the calculations and include the limitations stated by the source.
- The instructor has reviewed and approved the proposed topic before the team treats it as final.

References

1. **World Health Organization (WHO), Global Health Estimates.** https://www.who.int/data/global-health-estimates/
2. **WHO, Egypt country health profile.** https://www.who.int/data/gho/data/countries/country-details/GHO/egypt
3. **WHO Mortality Database.** https://platform.who.int/mortality/
4. **World Bank Data, Egypt.** https://data.worldbank.org/country/EG

Accessed: 16 October 2026. Confirm the source's current release and download metadata at the time the team retrieves the data.

Appendix A — Supplied CSV inspection

File inspected: Egypt_Synthetic_Mortality_Risk_250000 (1).csv. The filename explicitly contains "Synthetic". The file is a semicolon-delimited CSV with 25 columns and 250,000 records (row count checked from the file). The inspected fields include Record_ID, Baseline_Year, Governorate, Age, Sex, lifestyle/health-condition indicators, healthcare access, and Died_Within_12_Months. The naming and structure indicate a synthetic, person-level risk dataset; it is not evidence of actual registered deaths in Egypt.

Use policy: the file may be used only as clearly labelled synthetic practice data, if approved. It must not be merged with official mortality observations or used to claim real prevalence, mortality rates, governorate rankings, or causal effects.

Official-data starting point: WHO Global Health Estimates and the WHO Egypt profile. Confirm coverage, year, definitions, and data-quality caveats before final analysis.